

Algorithm 1.1: To check proper pairing of (), { }, and []

Algorithm **checkBalance**(expression)
 // Returns true if the parentheses, brackets, and braces in an
 // expression are paired correctly.

```

isBalanced = true
while ((isBalanced == true) and not at end of expression) {
    nextCharacter = next character in expression
    switch(nextCharacter) {
        case '(':
        case '[':
        case '{':
            push nextCharacter onto stack
            break

        case ')':
        case ']':
        case '}':
            if (stack is empty)
                isBalanced = false
            else {
                openDelimiter = pop the top entry from the stack
                isBalanced = true or false according to whether
                           openDelimiter and nextCharacter are
                           a pair of delimiters
            }
            break
    }
}

if (stack is not empty)
    isBalanced = false
return isBalanced

```